Database Lab : Disaster Recovery Replication Governance & Failover Quorum

Duration : 60 Min.

Scenario

As a Global Disaster Recovery Lead at NIIT, primary region telemetry indicates potential network partitioning across multi-engine database pairs. You must establish a disaster recovery governance table in MySQL, defining RPO/RTO thresholds, automatic failover eligibility, and split-brain fencing policies.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **MySQL Service:** `localhost:3306` | DB: `mysql` | User: `root` | Password: `MySQLPassword123!` | CLI: `mysql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **MySQL DR Table:** `mysql.<prefix>_dr_governance`

Task Objectives

Perform the following configuration steps directly via `mysql` CLI:

1. Create DR Governance Table

- In MySQL, create table `mysql.<prefix>_dr_governance` with columns:
- `dr_pair_id VARCHAR(50) PRIMARY KEY`
- `engine VARCHAR(30) NOT NULL`
- `rpo_sla_seconds INT NOT NULL`
- `rto_sla_seconds INT NOT NULL`
- `fencing_mechanism VARCHAR(50) NOT NULL`
- `auto_failover_eligible BOOLEAN NOT NULL`
- `certified_by VARCHAR(50) NOT NULL`

2. Populate Multi-Engine DR Specifications

- Insert the following 3 disaster recovery pair governance entries:
- `('DR-PG-PRIMARY', 'PostgreSQL 16', 10, 60, 'STONITH_WITNESS', TRUE, '<prefix>')`
- `('DR-MY-PRIMARY', 'MySQL 8.0', 10, 60, 'RAFT_QUORUM', TRUE, '<prefix>')`
- `('DR-MS-PRIMARY', 'SQL Server', 10, 60, 'CLUSTER_MAJORITY', TRUE, '<prefix>')`

3. Verify DR Policies Query

- Run `SELECT * FROM mysql.<prefix>_dr_governance WHERE auto_failover_eligible = TRUE;`.

Verification

Once completed, run the verification script:

```
python run.py
```

The driver connects to MySQL on `localhost:3306` and validates the DR governance table.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	MySQL DR Governance Table (_dr_governance) Creation	3 Marks
TC2	Multi-Engine DR Specification Population (Postgres, MySQL, MSSQL)	3 Marks
TC3	RPO (<10s) and RTO (<60s) SLA Standards Enforcement	3 Marks
TC4	Fencing Mechanism (STONITH / Quorum) Policy Definition	3 Marks
TC5	Automatic Failover Eligibility Rules Specification	2 Marks
TC6	Split-Brain Prevention Controls Verification	2 Marks
TC7	Certified By Prefix Integrity Check	2 Marks
TC8	End-to-End DR Replication Governance Sign-Off	2 Marks

Total Score: 20 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- All commands must be run in the live mysql console.